

Project 1:
Supervised Learning
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T-504-ITMAL Machine Learning

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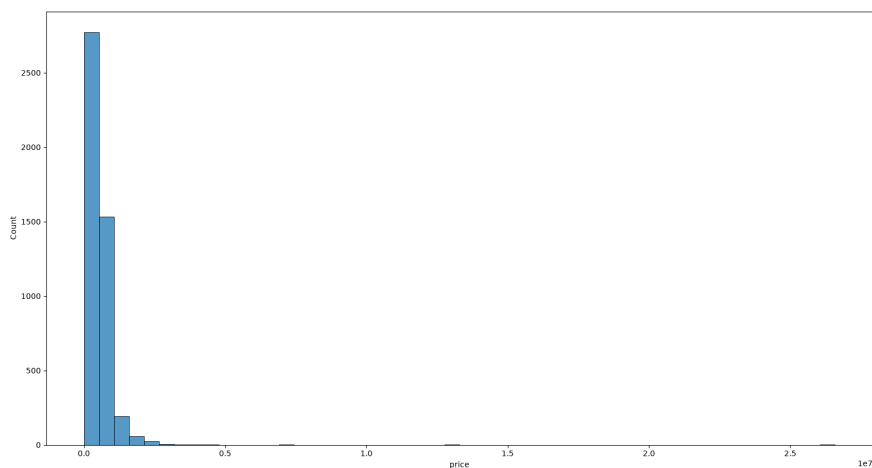
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1 Introduction

The dataset that I chose is the House Price Prediction dataset, I want to create a model that can effectively predict the price of a house based on its attributes. The dataset I chose gives us a plethora of features to work with. There's a total of 4,600 houses in the data set, each holding the square footage, number of bathrooms, number of bedrooms, how many floors, the condition it is in, the date it was sold, whether or not it has a waterfront and the rating of the view. The benchmark I'm going to be using is a normal Linear Regression, which is a common baseline for datasets like this.

A linear regression requires a dataset of only numerical values; the dataset that I chose has some string values in it, which can't be used in a linear regression. The problematic values here are "date", "street", "city" and "statezip". We can thankfully just drop the street attribute, as every single one of the values in it is unique. We can split the date into 2 attributes (day and month), the year is intentionally dropped here as the entire data range spans over less than a year, so it would actively hurt some models. For the "city" attribute, we can use a "one-hot" method on it, which turns each category into a boolean value, the only drawback here is that it shoots the dimensionality of the dataset up by quite a bit, from 18 to 60. The most

problematic of the string values is the "statezip" value. We can use a one-hot method on this as well, but this attribute shoots the dimensionality up even higher than the city attribute, from 60 to 136. Which is an increase of 62 features, vs. the increase of 42 with the city feature. The resulting Linear Regression gave us an accuracy of 30% with training data, and an accuracy of 60% with testing data. I also calculated the standard-deviations of the price to be \$427,379.5 for the training data, and \$927,053.6 for the testing data. This is also an extraordinary range, and it clearly hints that the dataset has a few outliers. I made a graph for the prices so I could see if this was the case:



There's a couple clear outliers in the data, a house that costs a whopping \$26,000,000 million Dollars, as well as some others between that, while most of the houses lie in the \$300k – 400k Dollar range. This graph unfortunately does not show all the houses with a price of \$0,

but there's a total of 49 properties with a listed price 0, this heavily skews the standard-deviation and hurts the model performance. For the 0 values, we can just remove the values, since 0 does not reflect the actual value of the property (low bid or perhaps given away). For the extreme values, we can take the logarithm of them, this will scale everything down to the same range. The resulting score of the regression model is 0.805 for the training data, and 0.774 for the testing data. I also calculated a RMSE of 0.269 and a MAE of 0.165 in the logarithmic scale, for the actual Dollar value, I got a MAE of \$110,188.80 and a RMSE of \$486,176.91. This may look like a very significant difference, but the RMSE jumps up more for higher values in the data than the MAE does, considering the biggest values present in our data is $> 10\times$ as much as the average, it makes a lot of sense, and we can't really do much about that, outside of removing the extreme values, which I can't really justify, as they do actually represent the values of the houses, different from the zero values. Considering everything, these will be our benchmarks.

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